Predicting the Final Size of an Epidemic using Numerical Methods

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Abstract

We aim to answer what fraction of the population will eventually be infected if an infectious disease is introduced into a population modeled by SIR. The SIR equations have no closed form solution, so we implement two numerical methods, Newton's Method and Fixed Point Iteration, to approximate the solution. To assess the convergence of our approximations, we estimate the rate of convergence of each method empirically and using convergence theory.

1 Introduction

The SIR model can be used to investigate the outcome of introducing a disease into a susceptible population. If we fix the population size, we can divide the population into three groups:

- $S(t)$: the fraction of susceptible individuals who can contract the disease.
- $I(t)$: the fraction of infected individuals who can transmit the disease.
- $R(t)$: the fraction of recovered individuals who are no longer infectious.

The SIR model can be described by a system of ordinary differential equations. If we let $\beta > 0$ denote the transmission rate and $\gamma > 0$ denote the recovery rate, then β/γ yields the average number of new infections caused by a single infectious person in a fully susceptible population. We call this the basic reproduction number and denote it by R_0. If $R_0 > 1$, the disease can spread rapidly, resulting in an epidemic. If $R_0 < 1$, the disease is likely to die out, as each infected person spreads the disease to less than one other person on average. [1]

1.1 SIR Model

It can be shown that the rate at which individuals transition between the three categories is given by

$$\begin{aligned} \frac{dS}{dt} &= -\beta SI \\ \frac{dI}{dt} &= \beta SI - \gamma I \\ \frac{dR}{dt} &= \gamma I \end{aligned}$$

and if we suppose the disease is introduced by a very small proportion of an otherwise fully susceptible population, we can take $S(0) \approx 1$ and $R(0) \approx 0$. We can show the final size of the epidemic, or the total fraction of the population that is infected over the course of the epidemic, satisfies $z = 1 - e^{-R_0 z}$.

1.2 SIR Model with Vaccination

If a fraction v of the population is vaccinated before the disease is introduced, then the final size equation can be shown to be $z = (1 - v)(1 - e^{-R_0 z})$.

2 Numerical Methods

Because we cannot find a closed-form solution for z, we opt to employ numerical methods to approximate the non-trivial solution with an error tolerance of 10^{-12}. To employ numerical methods, we must first determine the interval of interest on which a solution for z should be found. Consider the solution $z = 1 - e^{-R_0 z}$ and subtract $1-e^{-R_0 z}$ from both sides. This converts our equation into a root finding problem: $0 = z-1+e^{-R_0 z}$. As we can see, the trivial solution is $z^* = 0$. To find a meaningful solution, we need $z \in (0, 1)$ since it is a proportion of a population. When $R_0 = 2.5$, we can show that for $a = 0.1$, we have $f(a) = -0.1212$. For $b = 1$, we have $f(b) = 0.0821$. Because f is continuous and differentiable on $[a, b]$ and $0 \in [f(a), f(b)]$, we can apply the Intermediate Value Theorem to say that there must exist a value $z^* \in (0.1, 1)$ such that $f(z^*) = 0$.

Similarly, if we consider the function $g(z) = 1 - e^{-R_0 z}$, the solution $z = 1 - e^{-R_0 z}$ is a fixed point of g. The sequence generated by Fixed Point Iteration will converge to a unique fixed point in an interval $[a, b]$ if $g(z) \in [a, b]$ $\forall z \in [a, b]$ and $|g'(z)| \leq k$ for some $0 < k < 1$. It should be noted that for $R_0 = 2.5$, $g(z) = 1-e^{-2.5z}$ and $g'(z) = 2.5e^{-2.5z}$. The derivative of g is positive for all values of z, but we must select a tighter interval than we did for Newton's method. We expect the fixed point to occur in $[0.1, 1]$ as we showed for Newton's Method, but $|g'(0.1)| > 1$, so g is not contracting on this interval. Since $g''(z) = -6.25e^{-2.5z} < 0$ $\forall z$, $g'(z)$ is strictly decreasing. Since $g'(z) > 0$, it suffices to find a point where $g'(z) < 1$. Take $a = 0.4$. Then, $g'(a) \approx 0.92 < 1$, so $|g'(z)| < g'(a) < 1$ on $[0.4, 1]$. Since g is a contraction on this interval and $g([0.4, 1]) = [0.6321, 0.9179] \subseteq [0.4, 1]$, Fixed Point Iteration will converge to a unique fixed point here.

2.1 Newton's Method

Newton's Method approximates the zero z^* of the function f by repeatedly following the tangent line of f to where it crosses the x-axis. Given an initial approximation p_0, Newton's Method generates the sequence

$$p_{n+1} = p_n - \frac{f(p_n)}{f'(p_n)} \text{ for } n \in \mathbb{N} \quad .$$

2.2 Fixed Point Iteration

Fixed Point Iteration approximates a fixed point z^* of a function g, such that $g(z^*) = z^*$. We use the Fixed Point function $g(z) = 1 - e^{-R_0 z}$. Given an initial approximation p_0, Fixed Point Iteration generates the sequence

$$p_{n+1} = g(p_n) \text{ for } n \in \mathbb{N}$$

2.3 Order of Convergence

Both Newton's Method and Fixed Point Iteration are expected to converge on our chosen interval. We measure the Empirical Order of Convergence α_n at each iteration to estimate the order of convergence of the method. We calculate:

$$\alpha_k = \frac{\ln(|p_{k+2} - z^*|/|p_{k+1} - z^*|)}{\ln(|p_{k+1} - z^*|/|p_k - z^*|)}$$

3 Implementation

We implement Newton's Method and Fixed point iteration in Python, where we define functions for both methods using the above criteria. Newton's Method takes the root finding function $f(z) = z - 1 + e^{-R_0 z}$ and its derivative $f'(z) = 1 - R_0 e^{-R_0 z}$ as inputs, for which we define separate functions. It also takes an approximation tolerance and a maximum number of iterations, which are two stopping criteria for the method. Similarly, Fixed Point Iteration takes the fixed point function $g(z) = 1 - e^{-R_0 z}$ as an input, as well as an approximation tolerance and a maximum number of iterations. For our approximation tolerance we use the absolute error $|p_k - p_{k-1}| < \varepsilon$ and set the error tolerance $\varepsilon = 10^{-12}$ for both methods. We set the maximum number of iterations $N = 1000$.

Both methods also take an initial approximation p_0. We initialize Newton's Method at $p_0 = 1$ and Fixed Point Iteration at $p_0 = 0.5$. We vary the basic reproduction number R_0 over some clinically significant values, and generate a sequence of approximations P_n, as well as calculate the Empirical Order of Convergence for each iteration. From convergence theory, we expect fixed point iteration to converge linearly when $|g'(z^*)| < k$ for some $0 < k \leq 1$. We find $g'(z) = R_0 e^{-R_0 z}$. This function evaluated at a given R_0 and resulting z^* tells us the asymptotic error constant λ. We compute the ratios of absolute error in each iteration relative to the fixed point: $\frac{|p_{k+1} - z^*|}{|p_k - z^*|}$. Convergence theory for Fixed Point Iteration says these error ratios are expected to approach $|g'(z^*)|$.

4 Results

We display the results of applying Newton's Method and Fixed Point Iteration to the unvaccinated model with varying basic reproduction numbers. In each table we display the number of iterations required to reach a solution within our error tolerance, the approximation at each iteration, and the empirical order of convergence calculated for each iteration. For the Fixed Point Iteration tables, we also include the absolute

error ratios. Both error calculations are truncated to 5 decimal points using chopping, as no valuable information is lost by not including all 13 decimal places.

4.1 Unvaccinated Population

4.1.1 Basic Reproduction $\mathbf{R_0 = 2.5}$

Running the methods with $R_0 = 2.5$, in the range of estimates for the 1918 influenza pandemic and for the ancestral strain of SARS-CoV-2, yields $z^* = 0.892644753609$. [2, 3]

Table 1: Newton's Method

Iteration	P_n	**EOC**
0	1.0000000000000	1.92380
1	0.8967208237817	1.99833
2	0.8926522926223	
3	0.8926447536353	
4	0.8926447536092	
5	0.8926447536092	

Table 2: Fixed Point Iteration

Iteration	P_n	**EOC**	$\frac{\lvert p_{n+1}-z^* \rvert}{\lvert p_n-z^* \rvert}$
0	0.5000000000000	1.38022	0.45626
1	0.7134952031398	1.14358	0.33856
2	0.8319910467509	1.04418	0.28980
3	0.8750669914418	1.01238	0.27437
4	0.8878218984849	1.00336	0.27001
5	0.8913425217904	1.00090	0.26882
6	0.8922946805286	1.00024	0.26850
7	0.8925507570286	1.00006	0.26841
8	0.8926195230797	1.00001	0.26839
9	0.8926379818214	1.00000	0.26839
10	0.8926429361264	1.00000	0.26838
11	0.8926442658173	1.00000	0.26838
12	0.8926446226916	1.00001	0.26838
13	0.8926447184725	1.00004	0.26838
14	0.8926447441789	1.00015	0.26836
15	0.8926447510782	1.00056	0.26831
16	0.8926447529299	1.00211	0.26811
17	0.8926447534269	1.00797	0.26736
18	0.8926447535603	1.03075	0.26457
19	0.8926447535961	1.13314	0.25396
20	0.8926447536057		0.21160
21	0.8926447536083		
22	0.8926447536090		

4.1.2 Basic Reproduction $\mathbf{R_0 = 1.3}$

Running the methods with $R_0 = 1.3$, in the range of estimates for seasonal influenza, yields $z^* = 0.4229699520613$. [4]

Table 3: Newton's Method

Iteration	P_n	**EOC**
0	1.0000000000000	1.34325
1	0.5779338173785	1.74505
2	0.4494717684207	1.96301
3	0.4241858440127	1.99904
4	0.4229728204280	
5	0.4229699520773	
6	0.4229699520613	
7	0.4229699520613	

Table 4: Fixed Point Iteration

Iteration	P_n	**EOC**	$\frac{\lvert p_{n+1}-z^* \rvert}{\lvert p_n-z^* \rvert}$
0	0.5000000000000	0.95810	0.71380
1	0.4779542232390	0.96977	0.72395
2	0.4627761885538	0.97795	0.73106
3	0.4520707165278	0.98379	0.73612
4	0.4443918122759	0.98802	0.73979
5	0.4388176355479	0.99111	0.74246
6	0.4347362972628	0.99339	0.74443
7	0.4317291847305	0.99507	0.74588
...	...	...	...
37	0.4229714899529	0.99999	0.75013
38	0.4229711056927	1.00000	0.75013
39	0.4229708174447	1.00000	0.75013
40	0.4229706012188	1.00000	0.75013
41	0.4229704390195	1.00000	0.75013
42	0.4229703173475	1.00000	0.75013
...	...	...	...
65	0.4229699525521	1.00220	0.74871
66	0.4229699524295	1.00294	0.74824
67	0.4229699523375	1.00394	0.74760
68	0.4229699522685	1.00528	0.74674
69	0.4229699522167	1.00709	0.74559
70	0.4229699521779	1.00954	0.74404
71	0.4229699521487	1.01288	0.74194
72	0.4229699521269	1.01749	0.73909
73	0.4229699521105	1.02388	0.73519
74	0.4229699520982	1.03289	0.72981
75	0.4229699520890	1.04588	0.72229
76	0.4229699520821	1.06527	0.71159
77	0.4229699520769	1.09543	0.69596
78	0.4229699520730	1.14632	0.67230
79	0.4229699520701	1.24414	0.63435
80	0.4229699520679	1.49606	0.56763
81	0.4229699520662		0.42862
82	0.4229699520650		
83	0.4229699520641		

4.1.3 Basic Reproduction $\mathbf{R_0 = 4}$

Running the methods with $R_0 = 4$, in the range of estimates for seasonal smallpox, yields $z^* = 0.9801725987182$. [5]

Table 5: Newton's Method

Iteration	P_n	**EOC**
0	1.0000000000000	1.98970
1	0.9802364315821	
2	0.9801725994201	
3	0.9801725987182	
4	0.9801725987182	

Table 6: Fixed Point Iteration

Iteration	P_n	**EOC**	$\frac{\lvert p_{n+1}-z^* \rvert}{\lvert p_n-z^* \rvert}$
0	0.5000000000000	1.61039	0.24055
1	0.8646647167634	1.09436	0.10081
2	0.9685280582143	1.00855	0.08118
3	0.9792272289261	1.00068	0.07945
4	0.9800974798726	1.00005	0.07932
5	0.9801666401771	1.00000	0.07931
6	0.9801721261430	1.00000	0.07930
7	0.9801725612384	1.00001	0.07930
8	0.9801725957457	1.00016	0.07930
9	0.9801725984825	1.00209	0.07927
10	0.9801725986995	1.02790	0.07885
11	0.9801725987167		0.07345
12	0.9801725987181		
13	0.9801725987182		

4.1.4 Basic Reproduction $\mathbf{R_0 = 15}$

Running the methods with $R_0 = 15$, in the range of estimates for the measles virus, yields $z^* = 0.9999996940963$. [6] The EOC calculations are omitted in Table 7 because Newton's Method converged too quickly, and did not yield any meaningful values.

Table 7: Newton's Method

Iteration	P_n	**EOC**
0	1.0000000000000	
1	0.9999996940963	
2	0.9999996940963	

Table 8: Fixed Point Iteration

Iteration	P_n	**EOC**	$\frac{\lvert p_{n+1}-z^* \rvert}{\lvert p_n-z^* \rvert}$
0	0.5000000000000	1.80506	0.00110
1	0.9994469156299	1.00054	4.6076×10^{-6}
2	0.9999996915493		4.5768×10^{-6}
3	0.9999996940963		
4	0.9999996940963		

4.1.5 Summary: Unvaccinated Model

Table 9: Summary of final size approximations and iteration counts for the unvaccinated model

R_0	**Newton** z^*	**Newton Iterations**	**FPI** z^*	**FPI Iterations**	$\lvert g'(z^*) \rvert$
1.3	0.4229699520613	7	0.4229699520641	83	0.750
2.5	0.8926447536092	5	0.8926447536090	22	0.268
4	0.9801725987182	4	0.9801725987182	13	0.079
15	0.9999996940963	2	0.9999996940963	4	4.588×10^{-6}

4.2 Vaccinated Population

We test a range of values for v and display the resulting final size approximation and the number of iterations it took to converge for Newton's Method and Fixed Point Iteration. In particular, we find that as the vaccination rate gets closer to the herd immunity threshold $1 - 1/R_0$, convergence slows considerably. This is because the asymptotic linear convergence constant $|g_v'(z^*)| = |(1-v)R_0 e^{-R_0 z}|$ approaches 1. This can be seen in Table 11, as the number of iterations grows to 802!

Table 10: Newton's Method

v	P_n	**n**
0	0.8926447536092	5
0.3	0.4988886600005	6
0.5	0.1856851017515	8
0.55	0.0961515695071	8
0.59	0.0198360611680	11

Table 11: Fixed Point Iteration

v	P_n	**n**	$\lvert g_v'(z^*)\rvert$
0	0.8926447536090	22	0.268
0.3	0.4988886600011	31	0.716
0.5	0.1856851017552	100	0.786
0.55	0.0961515695142	188	0.898
0.59	0.0198360612075	802	0.951

4.3 Analysis

We interpret the final size z^* as the proportion of the population that will be infected with the disease over the course of the epidemic given a basic reproduction number R_0. Specifically, when $R_0 = 2.5$, it is in the range of estimates for two global pandemics: The 1918 Influenza Pandemic and the COVID-19 Pandemic. [2] According to our model, with no mortality and no preventative measures, a disease with this reproduction rate should infect 89.3% of the population. Likewise, if this disease were introduced into the state of Georgia, approximately 9.8 million of the 11 million people in the population would be infected.

4.3.1 Order of Convergence

We find that for Newton's Method, the empirical order of convergence approaches 2. This is consistent with a quadratic convergence. This is what we would expect, as the root does not appear to be a repeated root and our initial approximation p_0 is sufficiently close to z^*. This is supported by the first derivative of f. Since it is non-zero when evaluated at the final approximation z^*, we expect Newton's Method to converge quadratically.

For fixed point iteration, we expect the resulting sequence to converge linearly. This is because the magnitude of $g'(z^*)$ does not exceed one when evaluated at any of the fixed points. We find that the empirical order of convergence does approach 1, but in table 2, 4, and 6, as the approximations become increasingly close, floating-point arithmetic results in round-off errors, which propagate throughout the rest

of the EOC iterations for each chart.

Similarly, the error ratios $\frac{|p_{n+1}-z^*|}{|p_n-z^*|}$ approach $g'(z^*)$, but as p_n becomes increasingly close to z^*, subtracting them leads to round-off errors, and the calculated error ratios drift away from the predicted value.

4.3.2 Sensitivity of Newton's Method to p_0

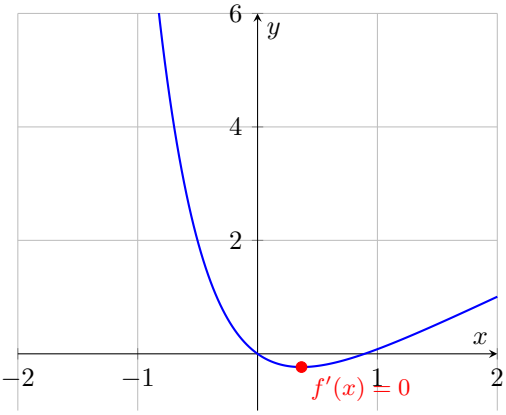
We test Newton's Method with $R_0 = 2.5$ while varying the initial approximations $p_0 \in (0, 1)$. According to Table 12, Newton's Method converges to the trivial solution for $p_0 \in (0, 0.366)$ and to the non-trivial solution for $p_0 \in (0.367, 1)$. This is what we would expect, since f reaches a local minimum at $x^* = 0.36652$. Taking p_0 to be the zero of f' would cause Newton's method to break, as P_1 would be undefined due to dividing by 0. Any value sufficiently close and to the right of x^* will yield a very small f'. So the first calculated approximation P_1 is very large. Interestingly, this does not significantly affect the number of iterations it takes for Newton's method to converge. Because $f(x)$ has the slant asymptote $y = x - 1$ when $x \to \infty$, for very large approximations P_1, Newton's method draws a tangent line $y \approx x + 1$, so the next approximation $P_2 \approx 1$. This can be seen exceptionally well when $p_0 = 0.37$. In Table 14, P_1 is much larger than the initial approximation, and $P_2 = 1$, accurate to at least 12 decimal points. The rest of the generated values match Table 1 exactly, where we had started with $p_0 = 1$. So even starting extremely close to the critical point of f' only added two iterations to our method.

The end behavior of f as x tends to negative infinity is not as conducive to quick convergence. When we select $p_0 < 0.367$, f' is negative. If we are sufficiently close to this critical point, P_1 is large and negative, and f' at each subsequent P_n is negative and large in magnitude. So, each iteration $p_{n+1} = p_n - \frac{f(p_n)}{f'(p_n)}$ only shifts our approximation to the right by a small step. Accordingly, in Table 13, a starting value pf $p_0 = 0.36$ requires 42 iterations to converge to 0.

Table 12: Newton's Method with $R_0 = 2.5$

$\mathbf{p_0}$	**Observed behavior**
0.10	Converges to $z = 0$
0.20	Converges to $z = 0$
0.30	Converges to $z = 0$
0.36	Converges to $z = 0$
0.37	Converges to $z^* \approx 0.892644753609$
0.50	Converges to $z^* \approx 0.892644753609$
0.75	Converges to $z^* \approx 0.892644753609$
1.00	Converges to $z^* \approx 0.892644753609$

Graph 1: $f(x) = x - 1 + e^{-2.5x}$



Graph 2: $f'(x) = 1 - 2.5e^{-2.5x}$

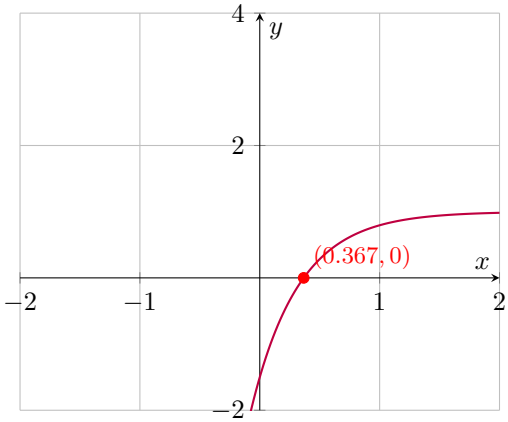


Table 13: Newton's Method $p_0 = 0.36$

n	P_n
0	0.3600000000000
1	-13.8526289321713
2	-13.4526289321713
3	-13.0526289321713
4	-12.6526289321714
5	-12.2526289321715
...	...
37	-0.0517742872012
38	-0.0049504983073
39	-0.0000504336164
40	-0.0000000052984
41	-0.0000000000000
42	-0.0000000000000

Table 14: Newton's Method $p_0 = 0.37$

n	P_n
0	0.3700000000000
1	27.2938153553561
2	1.0000000000000
3	0.8967208237817
4	0.8926522926223
5	0.8926447536353
6	0.8926447536092
7	0.8926447536092

4.3.3 Herd Immunity Threshold

We compare the final size approximation z^* to the herd immunity threshold for each tested value of R_0. The herd immunity threshold, $1 - \frac{1}{R_0}$, is the proportion of the population that must be immune for the epidemic to begin to decline. Note that according to Table 15 and 16, the final size of the epidemic surpasses the herd immunity threshold. This makes sense, as the epidemic begins to decline when the herd immunity threshold is reached, but the disease continues to spread, just at a slower rate, until reaching its final size. [7]

Table 15: Newton's Method

R_0	z^*	**Immunity Threshold**
2.5	0.8926447536092	0.6
1.3	0.4229699520613	0.23077
4	0.9801725987182	0.75
15	0.9999996940963	0.93333

Table 16: Fixed Point Iteration

R_0	z^*	**Total Iterations**	$\lvert g'(z)\rvert$
2.5	0.8926447536090	22	0.268
1.3	0.4229699520641	83	0.750
4	0.9801725987182	13	0.079
15	0.9999996940963	4	4.58×10^{-6}

4.4 Implementation Challenges

The empirical order of convergence and error ratio tables exhibited round-off error due to subtractive cancellation. I found a possible solution for the logarithm calculation through specialized library functions, but ultimately was not able to come up with a way to correct for the subtraction.

References

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